DevLab: I2C BMM150 Magnetometer Sensor

DEVLAB FACTOR

Mfr. Part #: UE0066

Product Reference Manual (V1.0)

Description

The BMM150 is a triaxial digital geomagnetic sensor optimized for low-power, high-precision applications. Engineered by Bosch Sensortec, it provides absolute magnetic field measurements along the X, Y, and Z axes, suitable for electronic compassing and inertial navigation tasks.

The device integrates seamlessly via I²C or SPI, ensuring compatibility with a wide range of host controllers, including Arduino, ESP32, and Raspberry Pi. Its compact form factor and minimal power profile make it ideal for space- and energy-constrained systems such as wearables, mobile robotics.

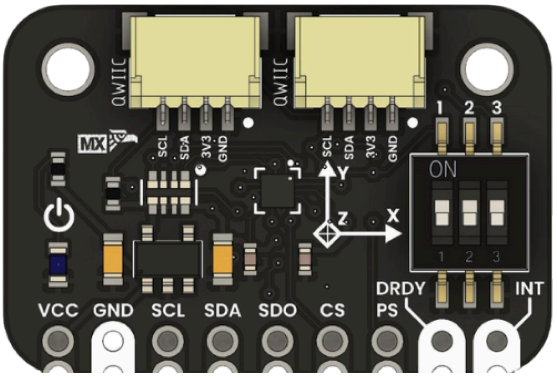
Built-in compensation algorithms correct soft-iron and hard-iron distortions, supporting reliable heading estimation in real-world environments. When used in conjunction with accelerometers and gyroscopes, the BMM150 enables robust 9DoF sensor fusion for advanced motion tracking and orientation systems.

DevLab format compatibility.

Simplicity and compatibility are the primary objectives of the DevLab form factor. It offers a board layout that is compact and optimized for serial communication, which includes I²C and SPI interfaces.

This format enables the establishment of rapid and dependable connections via a Protocol I²C connector that is entirely compliant with the Qwiic and STEMMA standards.

This design guarantees efficient prototyping, accessibility, and simplicity of integration across a variety of devices and modules.



Key Features

- **Axes:** 3 (X, Y, Z)
- **Measurement Range:** ±1300 µT
- **Resolution:** ~0.3 µT
- **Operating Voltage (VDD / VDDIO):** 3.3 V
- **Input Voltage via VCC Pin:** 3.6 – 6.0 V (via onboard LDO regulator)
- **Interfaces:** I²C, SPI
- **Operating Temperature Range:** Wide operating range suitable for various environments

Hardware Features

- Triaxial magnetoresistive sensing element
- Integrated analog front-end and digital signal processing
- Factory-calibrated sensor core
- Onboard LDO voltage regulator for flexible power supply
- DRDY (Data Ready) output for synchronized sampling
- INT (Programmable Interrupt) pin for event-driven operation
- Compact DevLab-compatible PCB with standardized pinout

Software Support

- **Arduino IDE**
  Official and community-supported libraries provide high-level APIs for **magnetic field measurement**, **heading calculation**, and **sensor configuration**, enabling rapid development on Arduino-compatible platforms.
- **MicroPython / CircuitPython**
  Bosch and community-developed drivers allow **lightweight scripting**, fast prototyping, and easy integration on platforms supporting MicroPython or CircuitPython.
- **Sensor Fusion Frameworks**
  Compatible with common **IMU and 9DoF fusion algorithms** when combined with accelerometers and gyroscopes.

Applications

- Electronic compass systems
- Inertial navigation systems (INS)
- Mobile robotics and autonomous platforms
- Wearable and portable devices
- Indoor navigation and orientation tracking
- Augmented and virtual reality systems
- Industrial magnetic field monitoring
- UAVs and mobile platforms requiring heading reference

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1 The Board

1.1 Accessories

- 1×9-pin 2.54 mm male header

2 Ratings

2.1 Recommended Operating Conditions

Symbol	Description	Min	Typ	Max	Unit
VCC	Input supply voltage (external input via VCC pin)	3.6	5.0	6.0	V
V3V3	Regulated 3.3 V output (AP2112K LDO)[1]	3.25	3.3	3.35	V
I3V3	3.3 V rail output current (total available)	300	350	600	mA
VIL	Low-level input voltage (I²C/SPI interface)	–0.3	–	0.99	V
VIH	High-level input voltage (I²C/SPI interface)	2.31	–	3.6	V
VOL	Low-level output voltage (at IOL = 3 mA)	–	–	0.4	V
VOH	High-level output voltage (at IOH = –3 mA)	2.4	–	3.3	V
ICC	Typical operating current (BME688 active mode)	–	12	18	mA
ISLEEP	Standby / sleep mode current	–	0.15	0.5	µA
RPU	I²C pull-up resistor to 3.3 V	4.7	–	10	kΩ
TOP	Operating temperature range	–40	–	+85	°C

[1] The AP2112K-3.3 linear regulator provides a stable **3.3 V output** for user applications, with a nominal tolerance of **±1.5%** (3.25 V – 3.35 V) at typical load conditions.
The maximum continuous output current is **up to 600 mA**, though sustained operation above **350 mA** may require thermal dissipation considerations depending on ambient temperature and PCB copper area.

3 Functional Overview

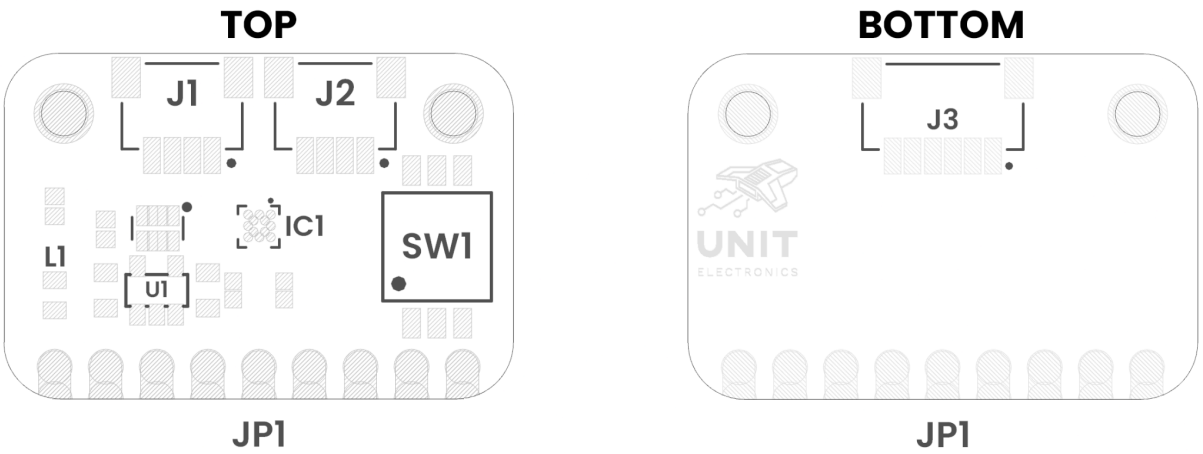
The **BMM150** is a **triaxial standalone geomagnetic sensor** that integrates the **magnetic sensing element and an application-specific integrated circuit (ASIC)** in a **wafer-level chip scale package (WLCSP)**. The device operates as a **slave** via standard **I²C or SPI digital interfaces**.

Designed for **consumer and embedded applications**, the BMM150 enables precise measurement of the **Earth's magnetic field along three perpendicular axes (X, Y, Z)**. The sensor is based on **Bosch Sensortec's proprietary FlipCore technology**, which provides optimized performance, stability, and low noise characteristics suitable for demanding **3-axis mobile applications**, including electronic compassing, navigation, and augmented reality systems.

An integrated **evaluation and signal-conditioning ASIC** converts the analog output of the geomagnetic sensing elements into calibrated digital data, which can be read via I²C or SPI. The device supports programmable operating modes, allowing the user to balance **performance, accuracy, and power consumption** according to application requirements.

The **ultra-small footprint** and **flat package design** make the BMM150 well suited for space-constrained designs such as mobile devices, wearables, and compact embedded systems.

3.2 Board Topology



Views of BMM150 Topology

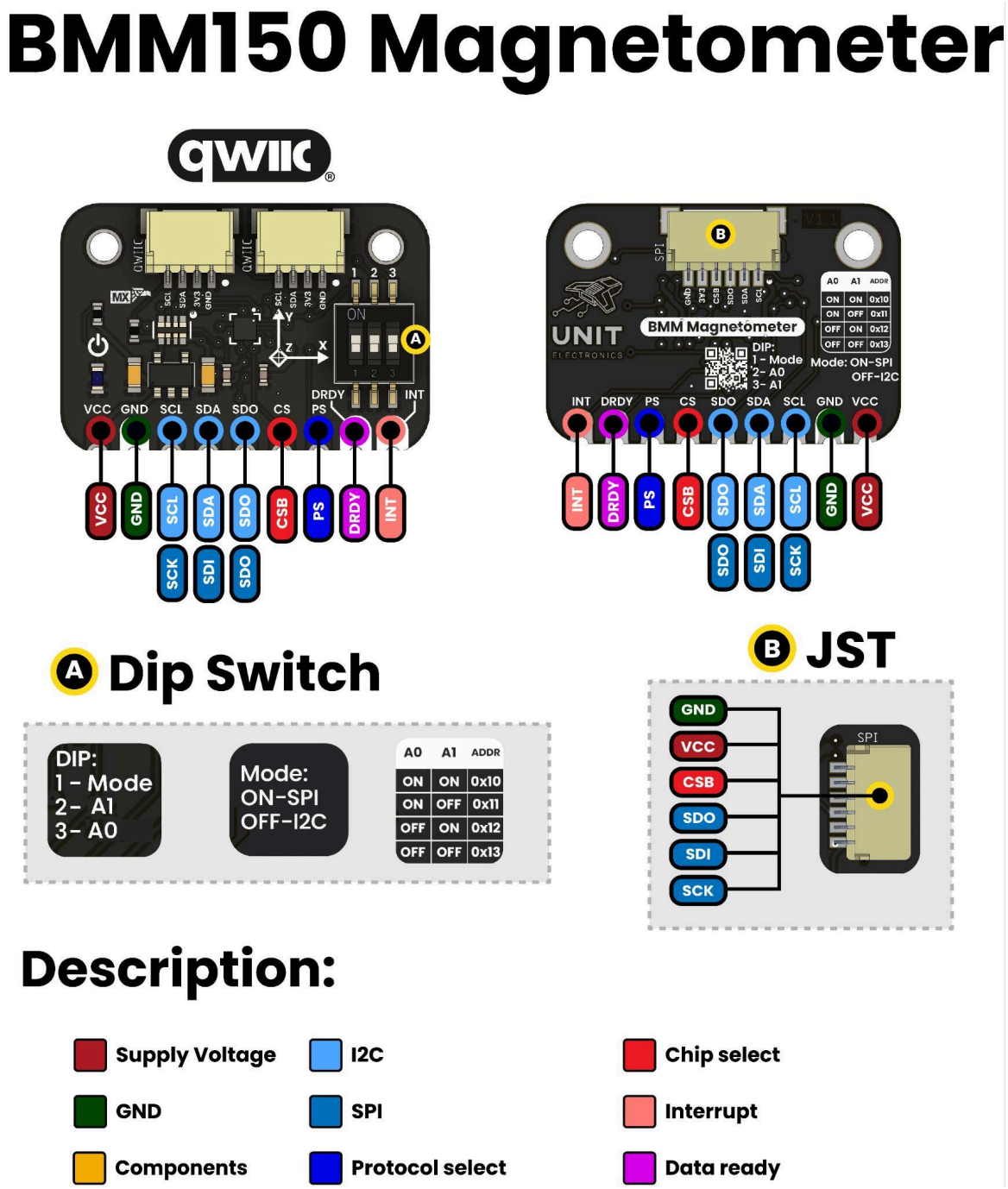
Views of I2C BMM150 Sensor Topology

Table 3.2.1 - Components Overview

Ref.	Description
IC1	BMM150 Magnetometer
U1	AP2112K 3V3 Regulator
L1	Power On LED
SW1	Dip Switch for Mode and Address Selector
J1	QWIIC Connector (JST 1mm for I2C)
J2	QWIIC Connector (JST 1mm for I2C)
J3	JST Connector 1mm Pitch for SPI
JP1	2.54mm Castellated Holes

4 Connectors & Pinouts

4.1 General Pinout



DevLab BMM150 Magnetometer Sensor Module *General Pinout*

4.2 Pinout General Description

Pin	Signal	Description
VCC	VCC	Power supply
GND	GND	Ground
SCL	SCL	I²C clock
SDA	SDA	I²C data
SDO	SDO / ADDR	SPI MISO / I²C address select
CS	CS	SPI chip-select (active LOW) / must be HIGH for I²C mode
PS	PS	Protocol select (LOW=I²C, HIGH=SPI)
DRDY	DRDY	Data-Ready flag (new data available)
INT	INT	Programmable interrupt output (e.g. threshold, flip-over)

5 Board Operation

This section describes the electrical operation, power requirements, communication interface, and typical usage of the BMM150 DevLab module. The board is designed to provide a simple and reliable hardware interface for integrating the BMM150 triaxial geomagnetic sensor into embedded systems using standard digital communication protocols.

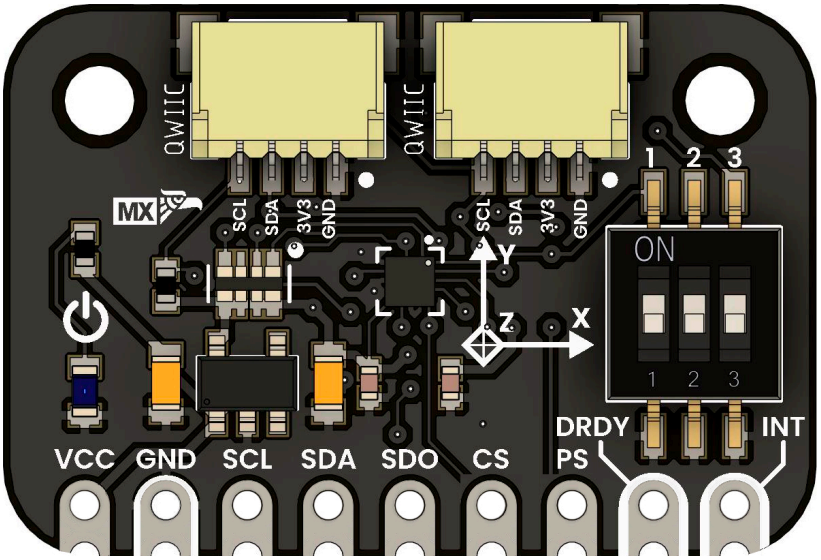
The DevLab implementation abstracts the complexity of the sensor's internal supply requirements by integrating onboard regulation and standardized connectors. This enables rapid prototyping while maintaining compatibility with a wide range of host controllers and operating environments.

5.1 I²C Address Selection

The BMM150 supports four selectable I²C slave addresses, enabling the use of multiple sensors on a single I²C bus.

The I²C address is selected using the DIP switches assigned to pins A0 and A1.

Name	A1 (Dipswitch 2)	A0 (Dipswitch 3)	I²C Address
ADDRESS_1	OFF	OFF	0x13
ADDRESS_2	OFF	ON	0x12
ADDRESS_3	ON	OFF	0x11
ADDRESS_4	ON	ON	0x10



DIP Switch 1 controls the PS sensor pin and selects the communication protocol mode.

- **ON position → Enables SPI mode**
- **OFF position → Enables I²C mode**

The user can keep all DIP switch interrupts in the **OFF** position and control the communication protocol mode through the **PS** pin header.
Additionally, the I²C address can be changed using the **CS** and **SDO** pin headers.

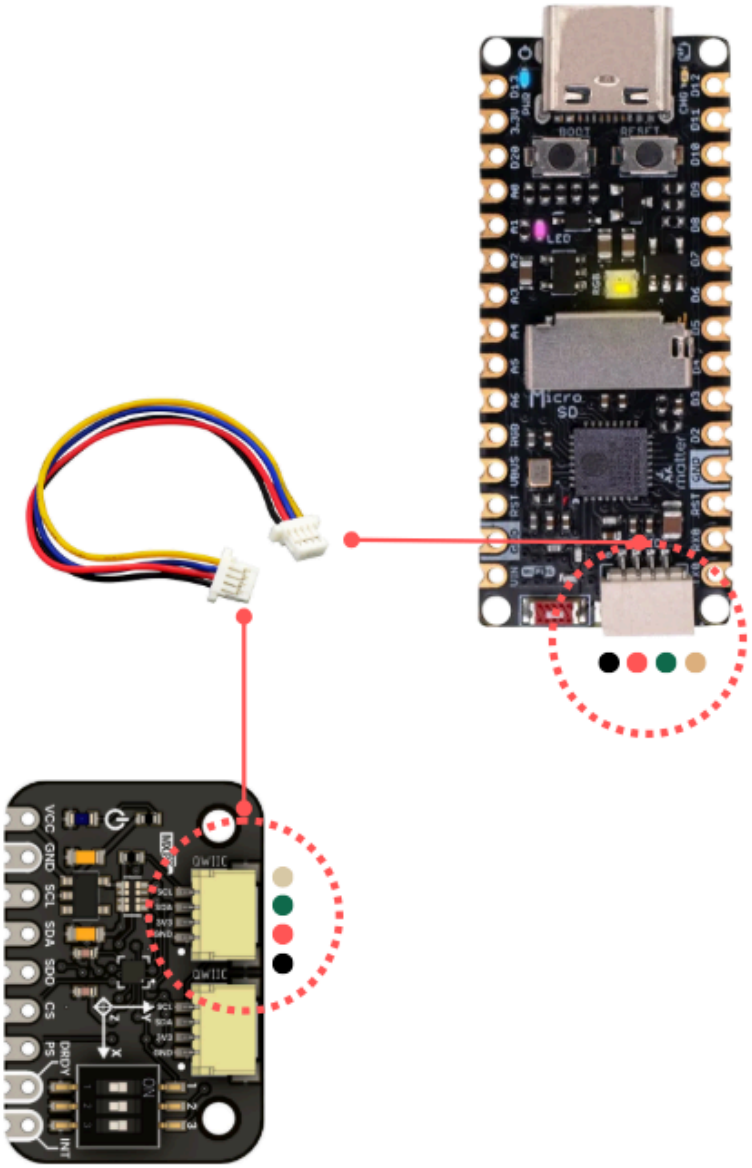
In **I²C mode (PS = 1)**:

SDO	CS	I²C Address
0	0	0x10
0	1	0x11
1	0	0x12
1	1	0x13

5.2 I²C – Arduino IDE

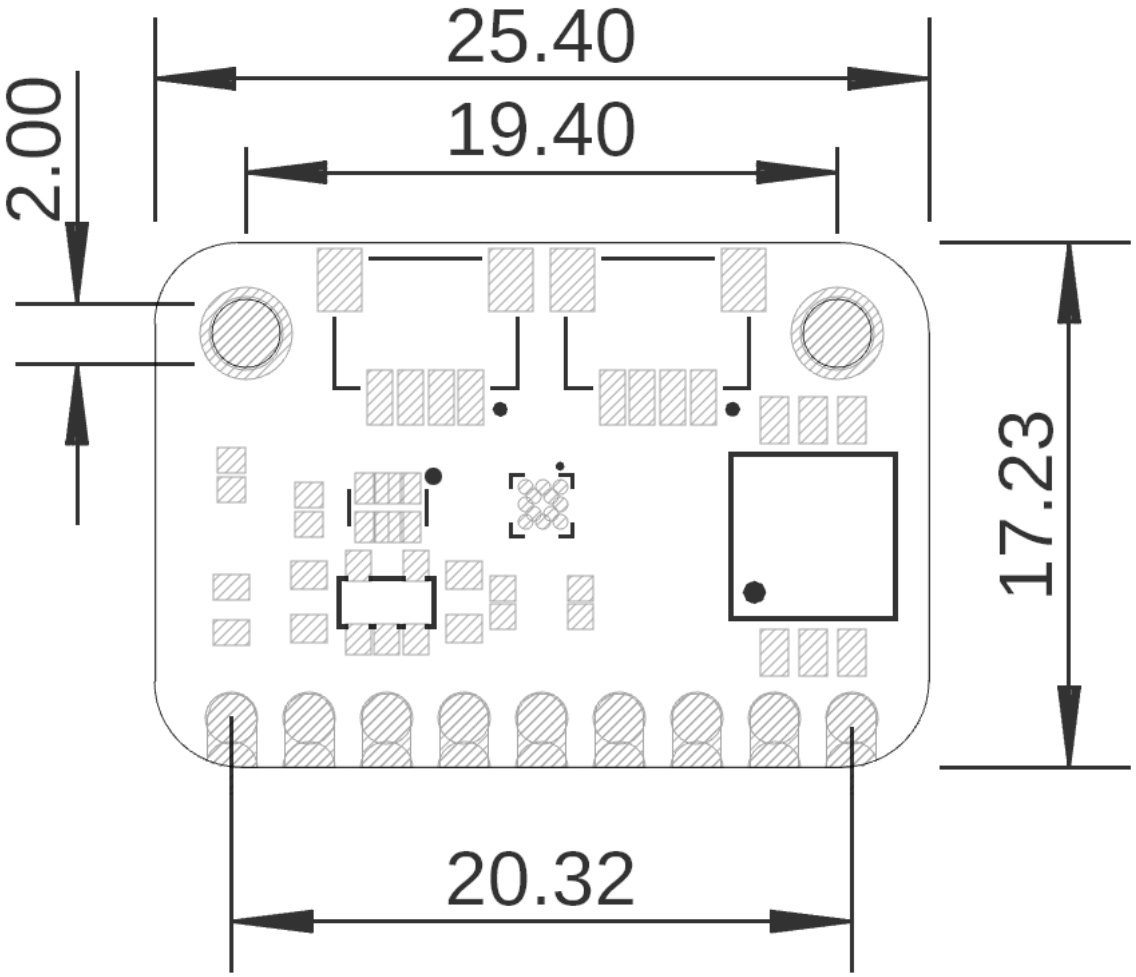
The following example demonstrates how to use the **BMM150** in **I²C mode** from the **Arduino IDE** with the **UNIT Pulsar ESP32-C6** development board. To view the measurement data, open the **Serial Monitor** and set the baud rate to **115200 bps**.

Pin	*Description*	*GPIO*
GND	Ground	
VCC	Power supply input	
SDA	I²C data line	6
SCL	I²C clock line	7



```
#include <Wire.h>
#include "DFRobot_BMM150.h"

// Crear un nuevo bus I2C en los pines GPIO 6 (SDA) y 7 (SCL)
TwoWire I2CBMM = TwoWire(1);

// Dirección I2C por defecto del BMM150 (depende del cableado de CS y SDO)
DFRobot_BMM150_I2C bmm150(&I2CBMM, I2C_ADDRESS_4);

void setup() {
  Serial.begin(115200);
  delay(1000);

  // Iniciar el nuevo bus I2C
  I2CBMM.begin(6, 7);  // SDA = 6, SCL = 7

  // Iniciar sensor
  while (bmm150.begin()) {
    Serial.println("bmm150 init failed, Please try again!");
    delay(1000);
  }
  Serial.println("bmm150 init success!");

  bmm150.setOperationMode(BMM150_POWERMODE_NORMAL);
  bmm150.setPresetMode(BMM150_PRESETMODE_HIGHACCURACY);
  bmm150.setRate(BMM150_DATA_RATE_10HZ);
  bmm150.setMeasurementXYZ();
}

void loop() {
  sBmm150MagData_t magData = bmm150.getGeomagneticData();
  Serial.print("mag x = "); Serial.print(magData.x); Serial.println(" uT");
  Serial.print("mag y = "); Serial.print(magData.y); Serial.println(" uT");
  Serial.print("mag z = "); Serial.print(magData.z); Serial.println(" uT");

  float compassDegree = bmm150.getCompassDegree();
  Serial.print("Compass angle (deg): ");
  Serial.println(compassDegree);
  Serial.println("------------------------------");

  delay(200);
}
```

6 Mechanical Information



Mechanical dimensions in millimeters

Mechanical dimensions in millimeters

7 Company Information

Company name	UNIT Electronics
Company website	https://uelectronics.com/
Company Address	Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX

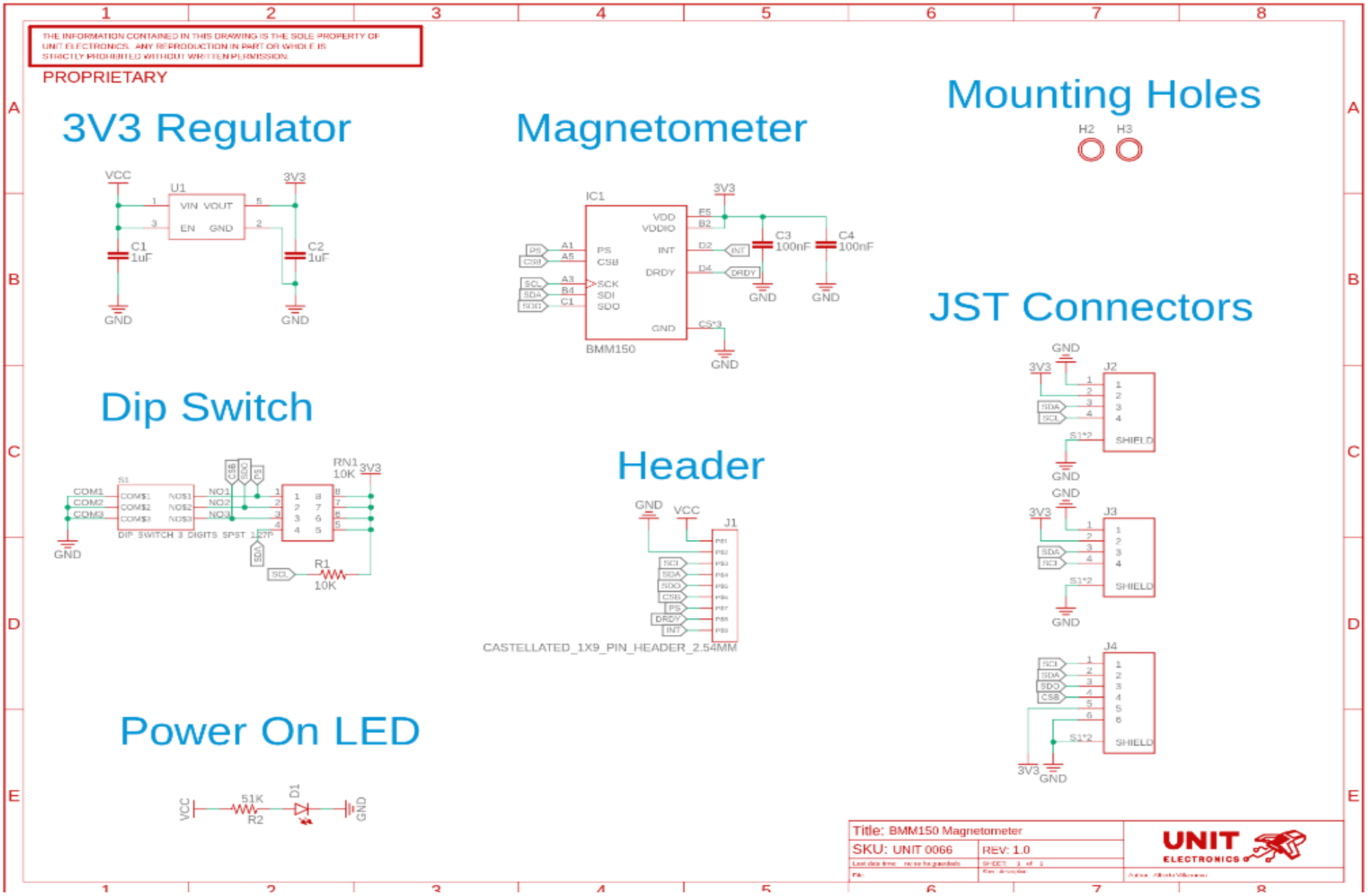
8 Reference Documentation

Ref	Link
Documentation	https://wiki.uelectronics.com/wiki/devlab-i2c-bmm150-magnetometer-sensor
Github	https://github.com/UNIT-Electronics-MX/unit_bmm150_magnetometer/tree/main
Thonny IDE	https://thonny.org/
Arduino IDE	https://www.arduino.cc/en/software
Visual Studio Code	https://code.visualstudio.com/download
DFRobot_BMM150 Library	https://github.com/UNIT-Electronics-MX/unit_bmm150_magnetometer/releases/download/1.0.0/DFRobot_BMM150-master.zip

9 Appendix

9.1 Schematic

(*https://github.com/UNIT-Electronics-MX/unit_bmm150_magnetometer/blob/main/hardware/unit_sch_V_0_0_1_ue0066_bmm150_magnetometro.pdf*)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PROPRIETARY
3V3 Regulator
VCC
U1
VIN VOUT
EN GND
3V3
C1 1uF
C2 1uF
GND
Magnetometer
IC1
VDD
VDDIO
PS
CSB
INT
DRDY
SCK
SDI
SDO
GND
BMM150
3V3
C3 100nF
C4 100nF
Mounting Holes
H2
H3
JST Connectors
J2
J3
J4
SHIELD
Dip Switch
RN1 10K
3V3
COM1
COM2
COM3
DIP SWITCH 3 DIGITS SPST
R1 10K
Header
J1
CASTELLATED_1X9_PIN_HEADER_2.54MM
Power On LED
51K R2
D1
Title: BMM150 Magnetometer
SKU: UNIT 0066
REV: 1.0
UNIT ELECTRONICS